

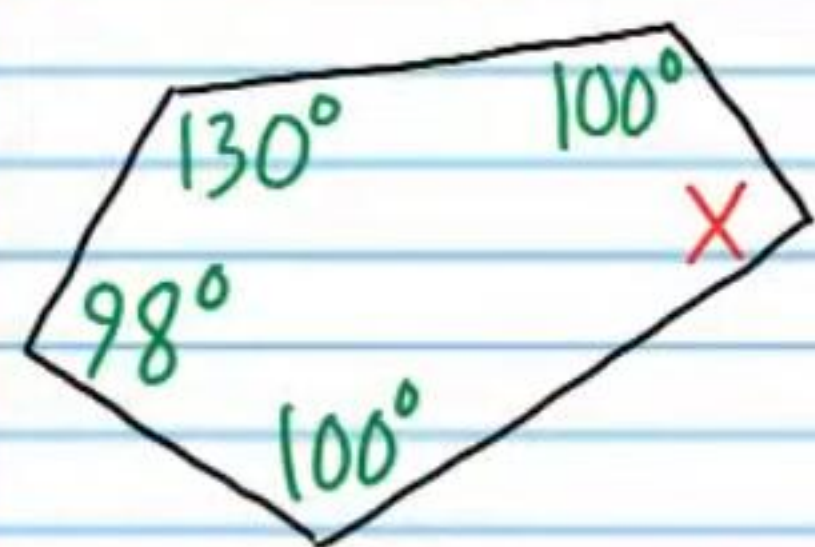
Practice Geometry Test IV (Version I)

Name
Date
Course

Instructions

- I) Write your name, the date, and the course title (i.e. Geometry).
- II) Show work when necessary.
- III) If a problem requires algebraic steps to solve, draw a rectangle around your final answer to distinguish it from the other work.
- IV) Each problem is worth four points for a total of 100 points.
- V) A calculator will be necessary for some problems on this test.

① Find x .



$$S = 180(n-2)$$

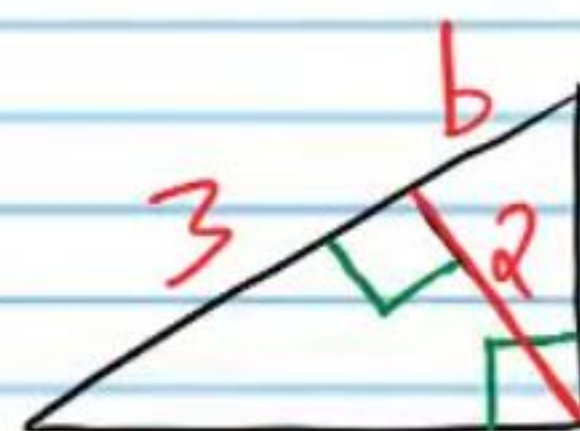
$$130 + 100 + 98 + 100 + x = 180(5-2)$$

$$428 + x = 180(3)$$

$$428 + x = 540$$

$$\boxed{x = 112^\circ}$$

② Find b .



$$h = \sqrt{xy}$$

$$2 = \sqrt{3b}$$

$$(2)^2 = (\sqrt{3b})^2$$

$$4 = 3b$$

$$\boxed{\frac{4}{3} = b}$$

③ Find the sum of the measures of the interior angles of the following polygon.

18-gon

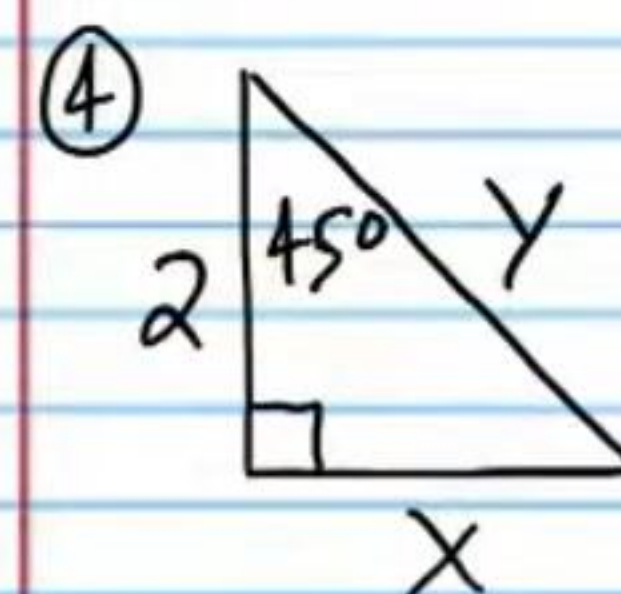
$$S = 180(n-2)$$

$$S = 180(18-2)$$

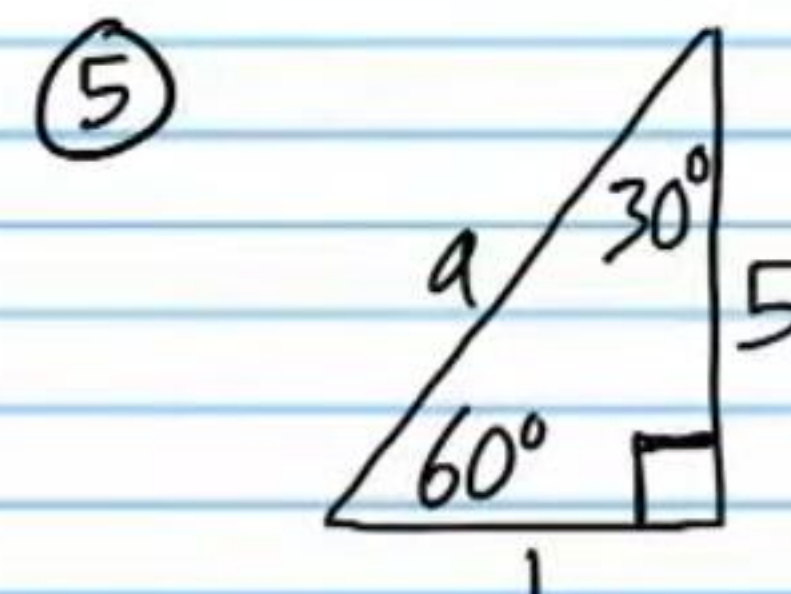
$$S = 180(16)$$

$$\boxed{S = 2,880^\circ}$$

Find the unknown values.



i) $x = \boxed{2}$ ii) $y = \boxed{2\sqrt{2}}$



i) $b = \boxed{\frac{5\sqrt{3}}{3}}$ ii) $a = \boxed{\frac{10\sqrt{3}}{3}}$

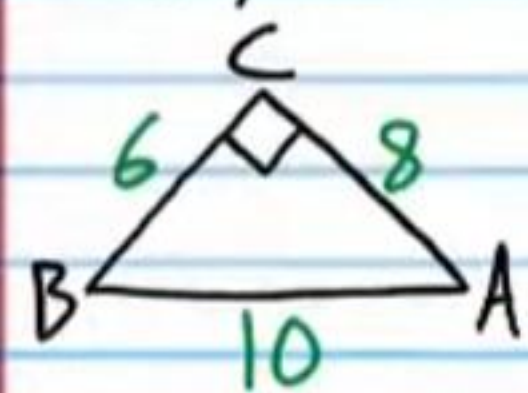
$$5 = w\sqrt{3} \quad a = 2w$$

$$\frac{5}{\sqrt{3}} = w \quad a = 2\left(\frac{5\sqrt{3}}{3}\right)$$

$$\frac{\sqrt{3} \cdot 5}{\sqrt{3} \cdot \sqrt{3}} = w \quad a = \boxed{\frac{10\sqrt{3}}{3}}$$

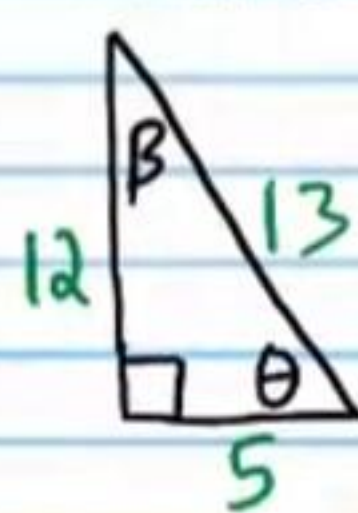
$$\frac{5\sqrt{3}}{3} = w$$

⑥ i) For the triangle below, find $\tan B$.



$$\tan B = \frac{8}{6} = \boxed{\frac{4}{3}}$$

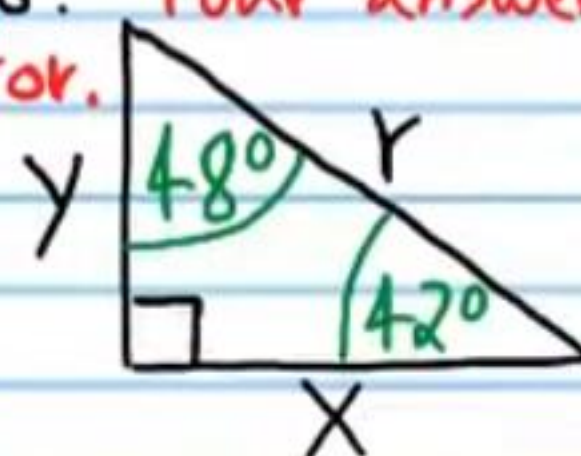
ii) For the triangle below, find $\cos B$.



$$\cos B = \frac{12}{13}$$

⑦ For the triangle at the right, find $\sin 48^\circ$. Your answer should have letters. Do not use a calculator.

$$\sin 48^\circ = \frac{x}{r}$$

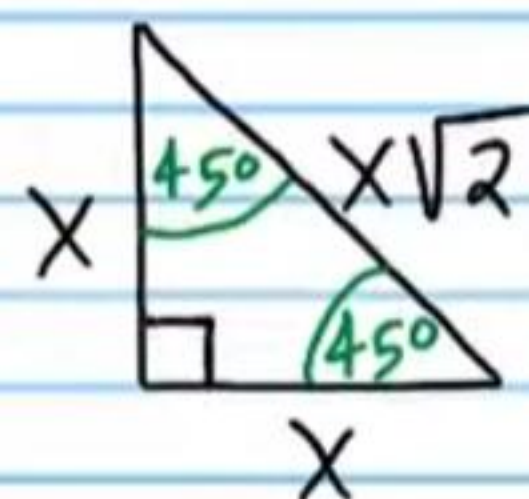


⑧ i) Use a calculator to find $\sin 87^\circ$. Round to the nearest thousandth.

$$\sin 87^\circ \approx \boxed{0.999}$$

Given the triangle below, find the following values without a calculator.

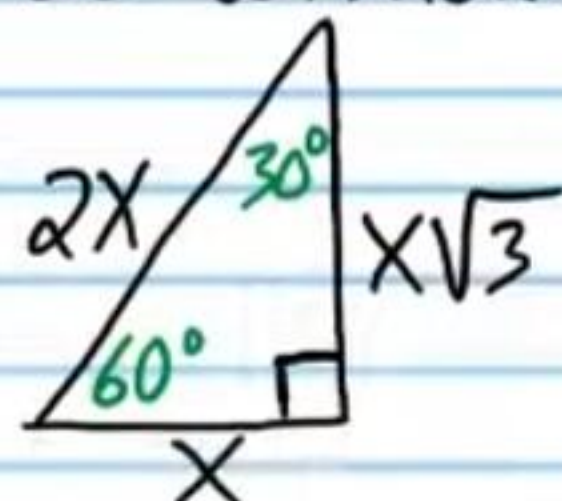
ii) $\sin 45^\circ = \frac{x}{x\sqrt{2}} = \frac{1}{\sqrt{2}} = \boxed{\frac{\sqrt{2}}{2}}$



iii) $\tan 45^\circ = \frac{x}{x} = \boxed{1}$

iv) Given the triangle below, find $\tan 60^\circ$ without a calculator.

$$\tan 60^\circ = \frac{x\sqrt{3}}{x} = \frac{\sqrt{3}}{1} = \boxed{\sqrt{3}}$$

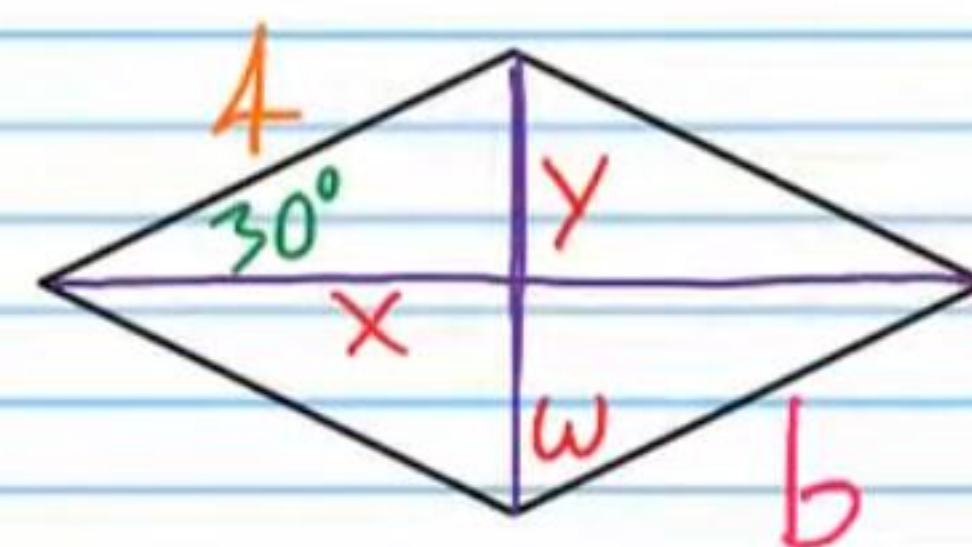


⑨ Find the unknowns in the rhombus below.

i) $x = \boxed{2\sqrt{3}}$

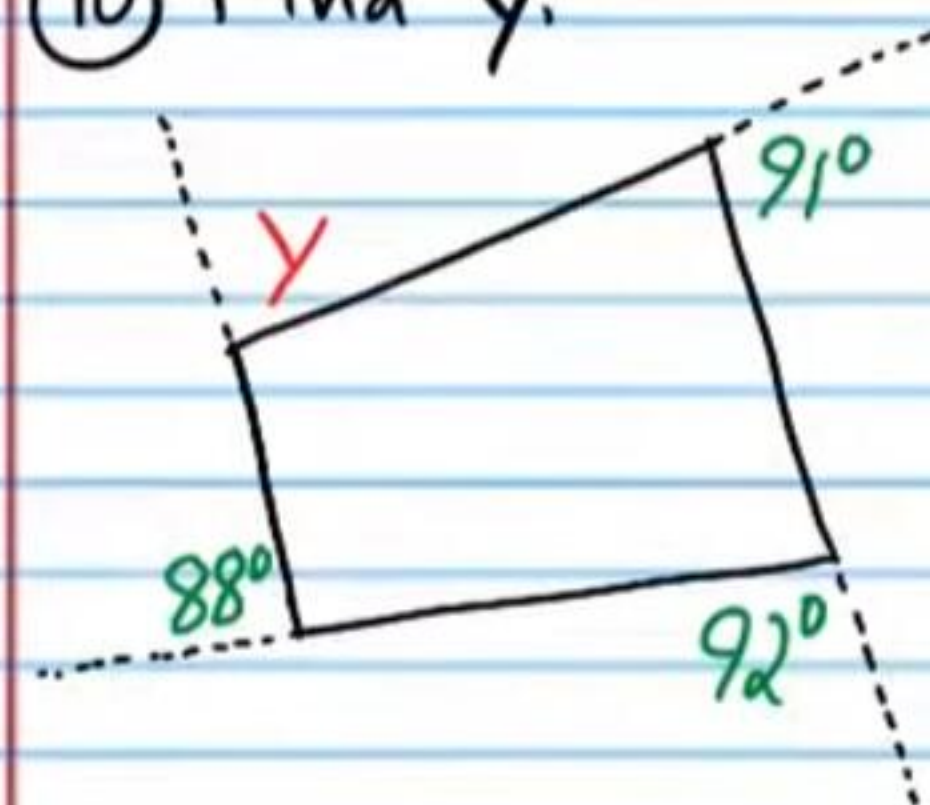
ii) $y = \frac{4}{2} = \boxed{2}$

iii) $b = \boxed{4}$



iv) $w = \boxed{60^\circ}$

⑩ Find y .



$$S = 360^\circ$$

$$y + 91 + 92 + 88 = 360$$

$$y + 271 = 360$$

$$\boxed{y = 89^\circ}$$

- ⑪ If the number below is the sum of the measures of the interior angles of a convex polygon, find the number of sides of the polygon.

$$1260^\circ \quad S = 180(n-2)$$

$$1260 = 180(n-2)$$

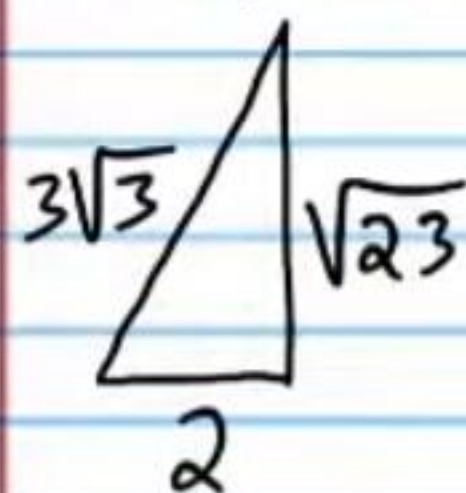
$$1260 = 180n - 360$$

$$1620 = 180n$$

$$9 = n$$

Nine sides

- ⑫ Is the following triangle an acute, obtuse, or right triangle? ⑬ Find the unknown values of the parallelogram below.



$$c^2 \stackrel{?}{=} a^2 + b^2$$

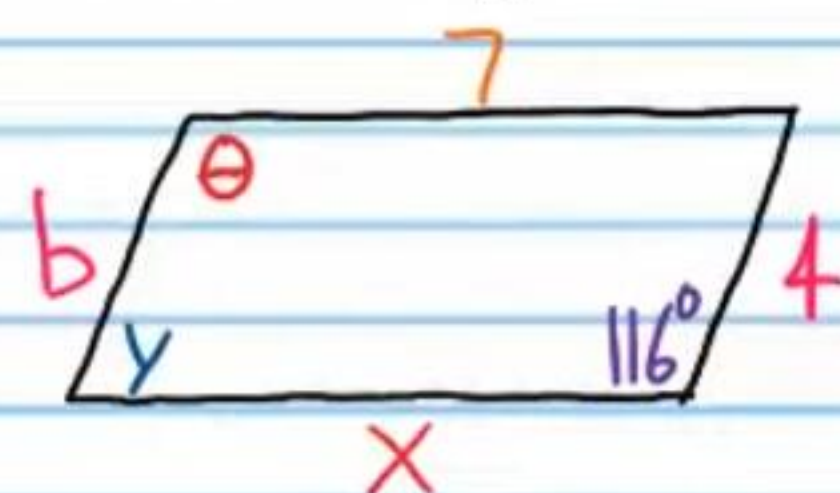
$$(3\sqrt{3})^2 \stackrel{?}{=} 2^2 + (\sqrt{3})^2$$

$$3\sqrt{3} \cdot 3\sqrt{3} \stackrel{?}{=} 4 + 3$$

$$9 \cdot 3 \stackrel{?}{=} 27$$

$$27 \stackrel{?}{=} 27$$

Right



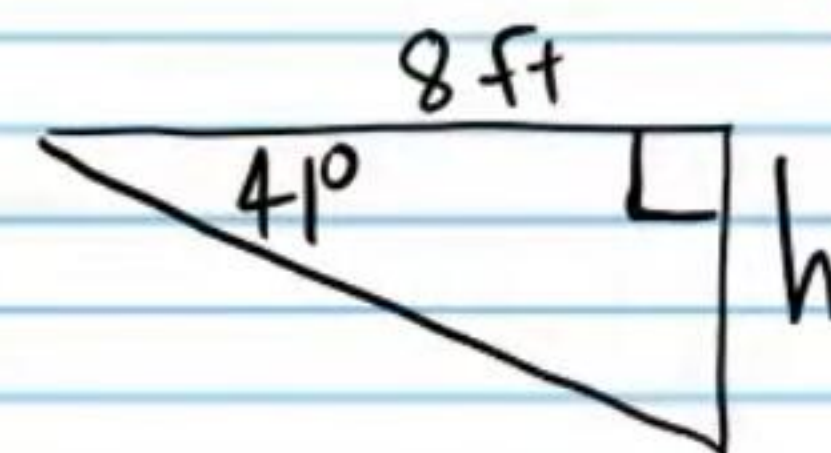
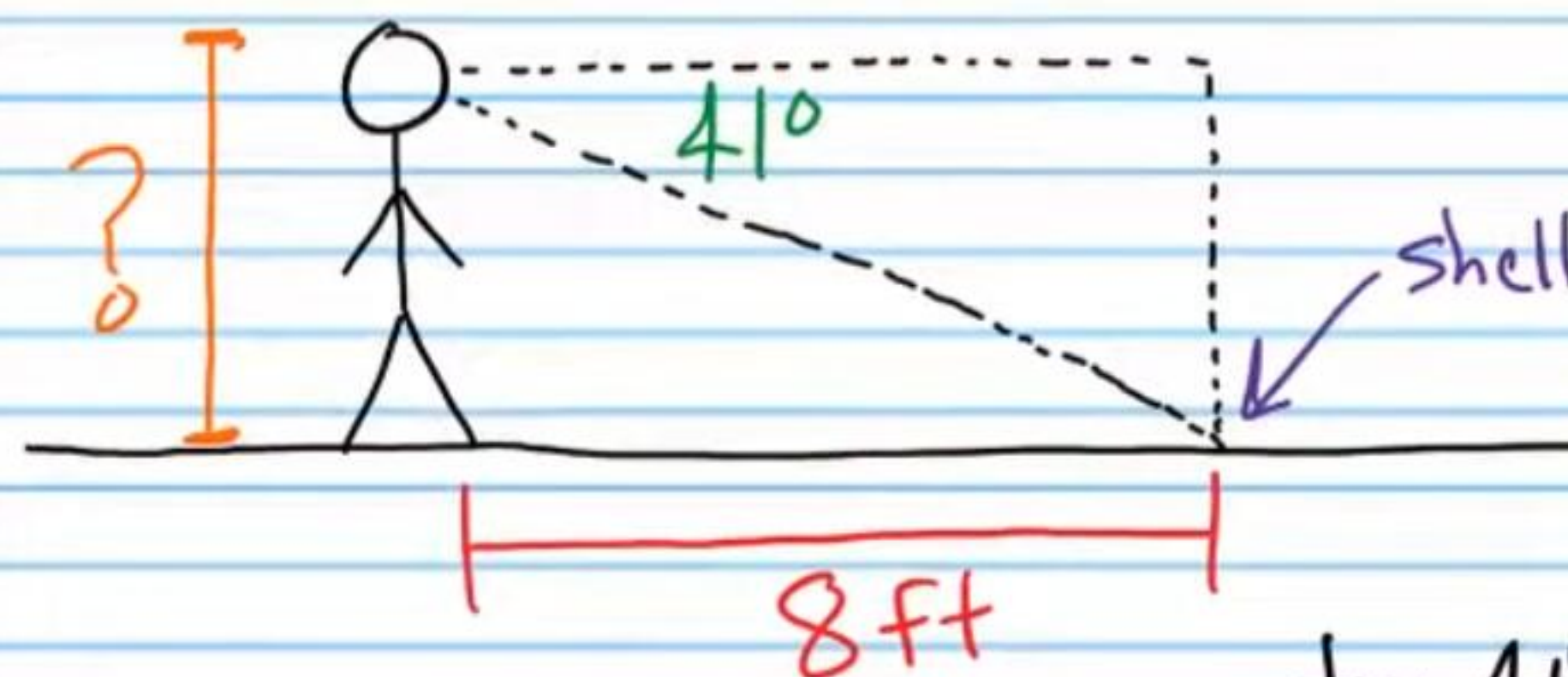
$$\text{i) } b = 4 \quad \text{ii) } x = 7$$

$$\text{iii) } \theta = 116^\circ \quad \text{iv) } y = 64^\circ$$

$$y + 116 = 180$$

$$y = 64^\circ$$

- ⑭ A woman looks down at a seashell on the beach. The angle of depression is 41° and she is standing 8 feet from the shell. How tall is she?



$$\tan 41^\circ = \frac{h}{8}$$

$$8 \cdot \tan 41^\circ = \frac{h}{8} \cdot 8$$

$$8 \tan 41^\circ = h$$

$$6.95 \text{ ft} \approx h$$

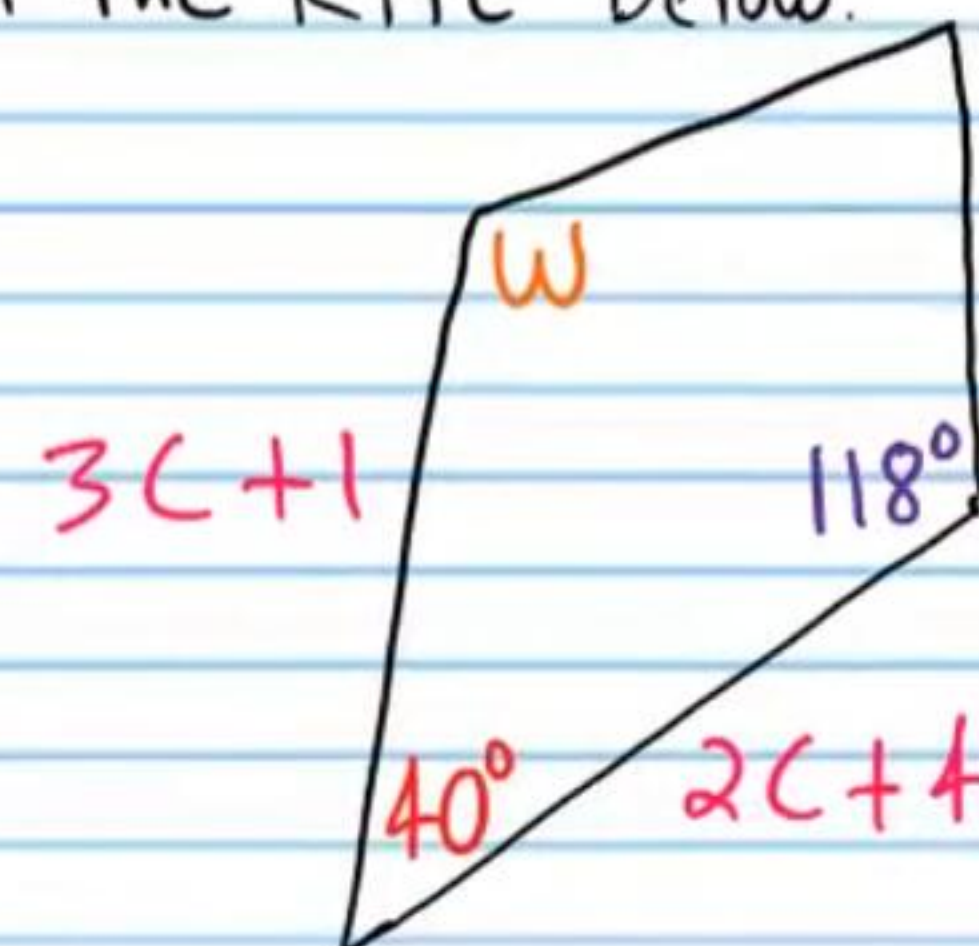
- ⑮ Find the unknowns in the kite below.

$$\text{i) } w = 118^\circ$$

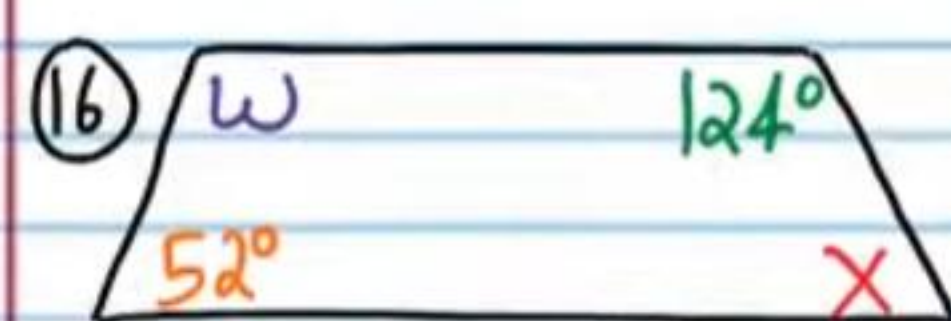
$$\text{ii) } c = 3$$

$$3c + 1 = 2c + 4$$

$$c = 3$$

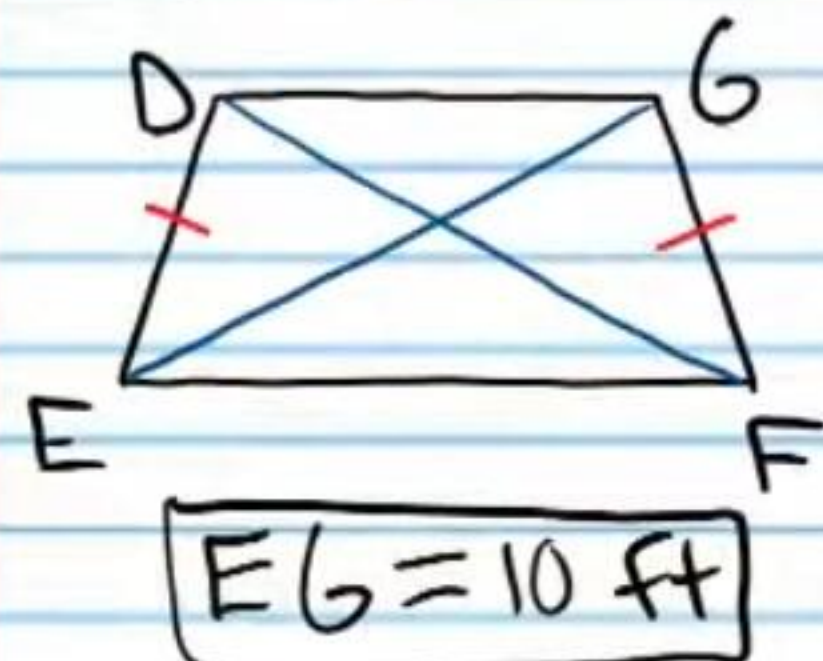


Find the unknowns in each trapezoid below.



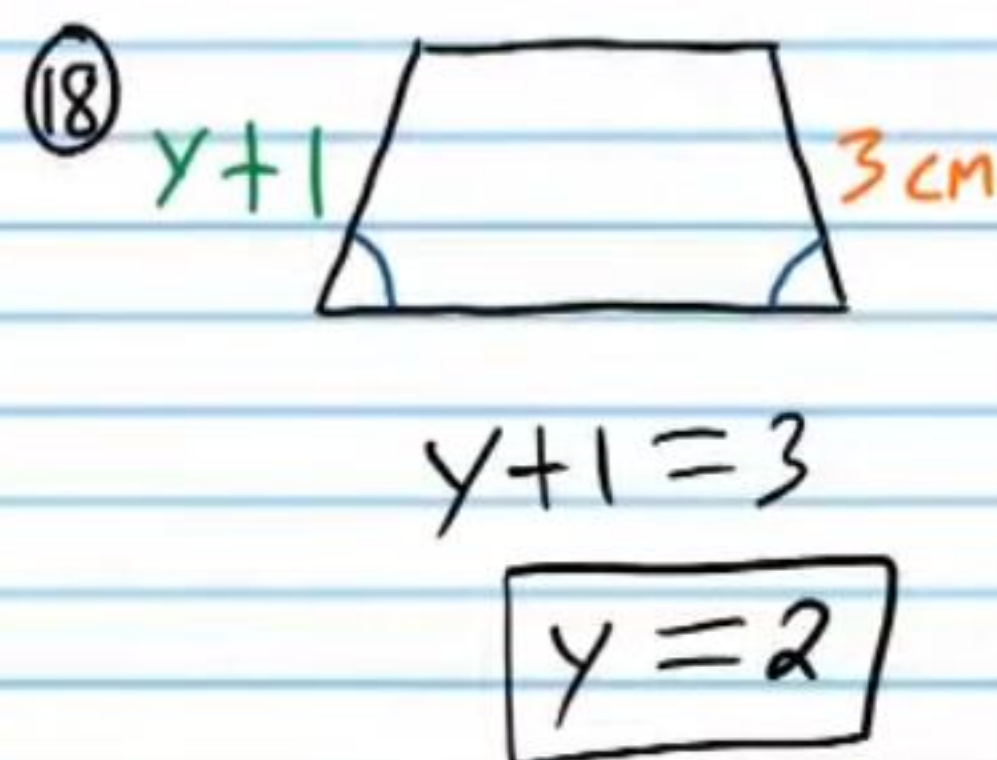
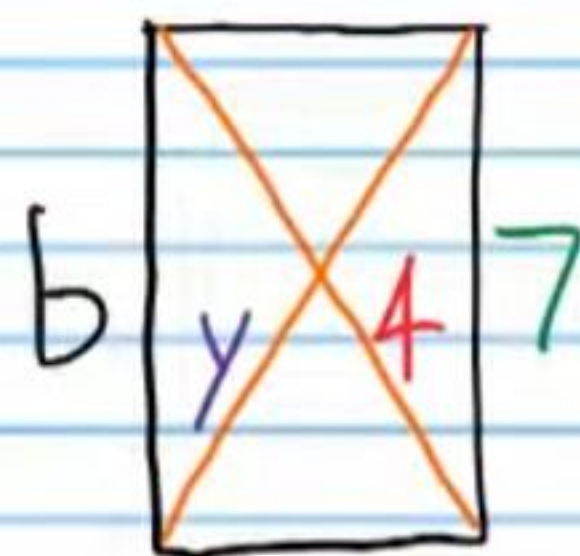
i) $x = \boxed{56^\circ}$ ii) $w = \boxed{128^\circ}$
 $x + 124 = 180$ $w + 52 = 180$
 $x = 56^\circ$ $w = 128^\circ$

①⑦ If $DF = 10$ ft, find EG .



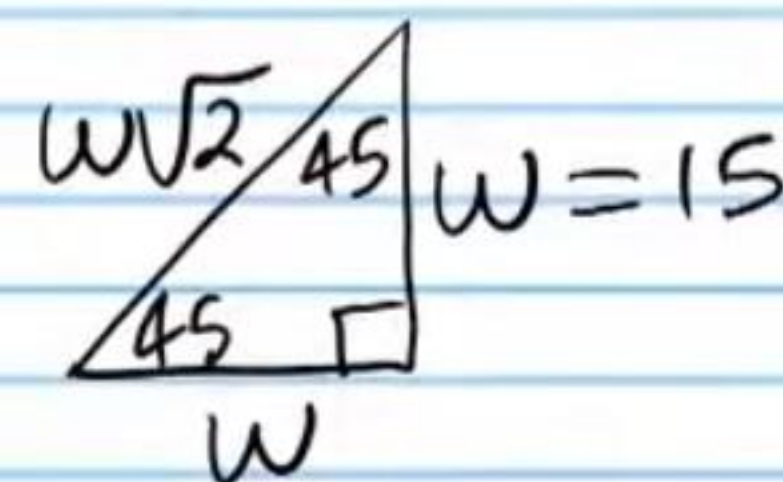
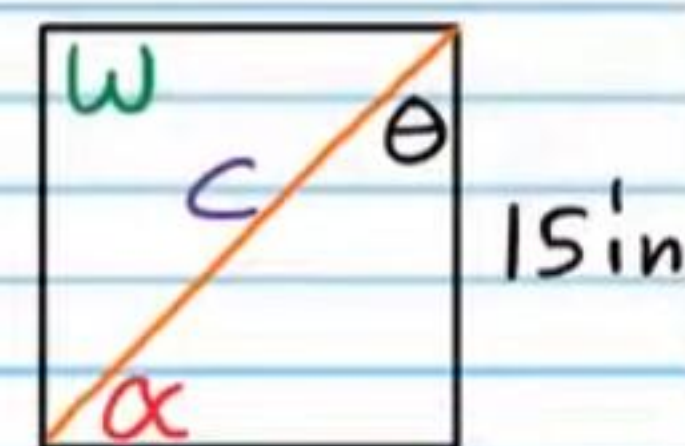
Find the unknowns in the rectangle below.

①⑨ i) $b = \boxed{7}$
 ii) $y = \boxed{4}$



②⑩ Find the unknowns in the square below.

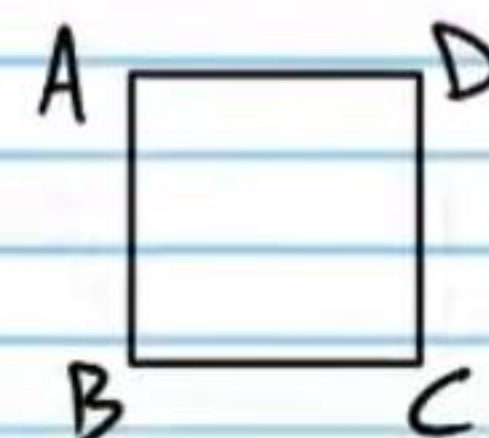
i) $w = \boxed{90^\circ}$ ii) $\alpha = \boxed{45^\circ}$
 iii) $\theta = \boxed{45^\circ}$ iv) $c = \boxed{15\sqrt{2} \text{ in}}$



Show that the following statements are true using two-column proofs. You might not use the same number of steps written in Roman numerals. You may use more or less.

②⑪ If the quadrilateral below is a square, then it is a rectangle.

Statement	Reason
I) ABCD is a square.	Given.
II) $\angle A, \angle B, \angle C, \& \angle D$ are right angles.	Def. of a $\square$.
III) ABCD is a rectangle.	Def. of a $\square$.

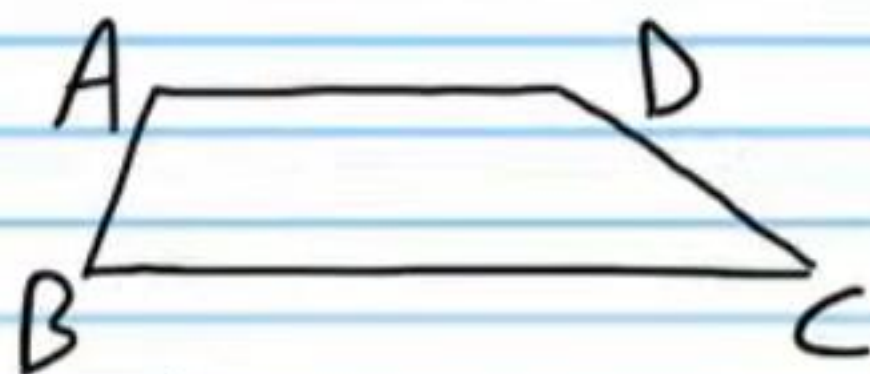


② If the figure below is a kite, then $\angle D \cong \angle B$. This is the Kite Congruent Angles Theorem.

Statement	Reason
I) ABCD is a kite.	Given.
II) $\overline{DA} \cong \overline{AB}$ & $\overline{DC} \cong \overline{BC}$	Def. of a kite.
III) $\overline{AC} \cong \overline{AC}$	Ref. Prop of $\cong$.
IV) $\triangle ADC \cong \triangle ABC$	SSS
V) $\angle D \cong \angle B$	CPCTC

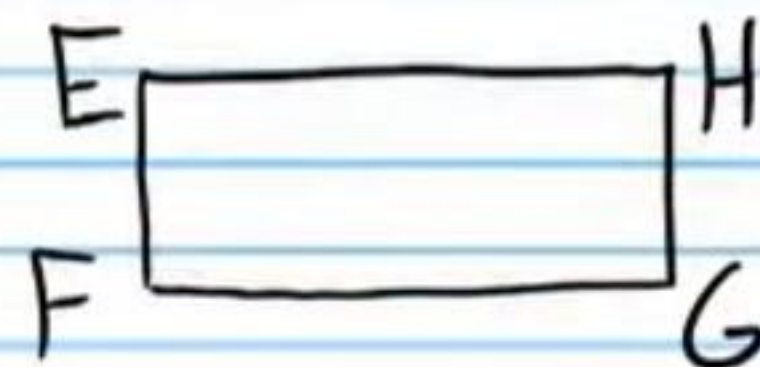


③ If the figure below is a trapezoid, then $\angle A$ and $\angle B$ are supplementary. Hint: You must use a transversal theorem.



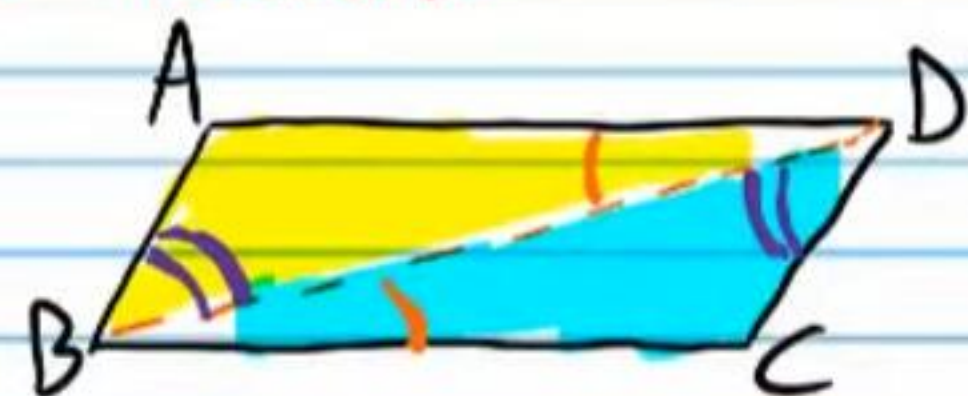
Statement	Reason
I) ABCD is a trapezoid.	Given
II) $\overline{AD} \parallel \overline{BC}$	Def. of a $\square$.
III) $\angle A$ & $\angle B$ are supplementary.	CIAT

④ If the quadrilateral EFGH below is a rectangle, then it is a parallelogram. Hint: You must use the Consecutive Interior Angles Converse Theorem.



Statement	Reason
I) EFGH is a rectangle.	Given.
II) $\angle E$, $\angle F$, and $\angle G$ are right $\angle$'s.	Def. of a $\square$.
III) $m\angle E = 90^\circ$, $m\angle F = 90^\circ$, and $m\angle G = 90^\circ$.	Def. of a rt $\angle$.
IV) $m\angle E + m\angle F = 90 + 90$ and $m\angle F + m\angle G = 90 + 90$.	Subs. Prop of $=$.
V) $m\angle E + m\angle F = 180$ and $m\angle F + m\angle G = 180$.	Simplification.
VI) $\angle E$ & $\angle F$ are supplementary and $\angle F$ & $\angle G$ are supplementary.	Def. of supp. $\angle$'s.
VII) $\overline{EH} \parallel \overline{FG}$ & $\overline{EF} \parallel \overline{HG}$	CIAT
VIII) EFGH is a parallelogram.	Def. of a $\square$.

25) If the quadrilateral below is a parallelogram, then $\overline{AB} \cong \overline{DC}$ and $\overline{AD} \cong \overline{BC}$. This is the Parallelogram Opposite Sides Theorem.



Statement	Reason
I) ABCD is a parallelogram.	Given.
II) $\overline{AB} \parallel \overline{DC}$	Def. of a $\square$.
III) $\angle ABD \cong \angle BDC$	AIAT
IV) $\overline{AD} \parallel \overline{BC}$	Def. of a $\square$.
V) $\angle ADB \cong \angle CBD$	AIAT
VI) $\overline{BD} \cong \overline{BD}$	Ref. Prop of $\cong$.
VII) $\triangle ABD \cong \triangle DCB$	ASA
VIII) $\overline{AB} \cong \overline{DC}$ & $\overline{AD} \cong \overline{BC}$	CPCTC
IX)	